BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 127 · CPU Performance and Embedded Systems

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does the Program Counter hold?

Q2. A = 1 and B = 0. What is A AND B?
A. 1
B. 2
C. 0
D. 10

Q3. “A NIC's purpose is keeping files when power is off.” Correct this. Explain why.

Q4. Marks must be integers from 0 to 100. Classify 50, 101 and text 'fifty'. Why does accepting 50 not prove it is the pupil's real mark?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

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I Do · Read new knowledge

Key vocabulary

Clock speed — The number of CPU cycles per second, commonly measured in GHz.

Number of cores — More cores can process more independent tasks concurrently.

Cache size — Larger cache can reduce slower memory accesses for frequently used data.

Embedded system — A dedicated computer designed into another device for a limited/specific purpose.

New learning

Clock speed: The number of CPU cycles per second, commonly measured in GHz.

Number of cores: More cores can process more independent tasks concurrently.

Cache size: Larger cache can reduce slower memory accesses for frequently used data.

Embedded system: A dedicated computer designed into another device for a limited/specific purpose.

Worked example

Compare CPU A: 2 GHz, two cores with CPU B: 3 GHz, four cores, with other factors unspecified.

1. B has more cycles per second.

2. B also has more cores for potentially parallel work.

3. It may perform better in suitable tasks, but other hardware/software factors matter.

4. Do not state that it must always run every program faster.

We Do · Practise with guidance

Guided 1

A washing machine has a controller that operates its cycle. Explain why it is an embedded system.

Worked method: It performs a specific function rather than general-purpose computing.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Calculate how many million cycles a 2 MHz clock completes in three seconds.

```
clockMHz = 2
seconds = 3
millions = ...
print(millions)
```

write

Question 1

State two CPU factors affecting performance. [2 marks]

Question 2

Explain why a larger cache may improve performance. [3 marks]

Question 3

Give an example of an embedded system and explain why it qualifies. [3 marks]

Exit Ticket · Answer before checking

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

More CPU cores can improve performance for tasks that run in parallel.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The address of the next instruction to be fetched.

Q2. C — 0

Q3. The interface that enables a device to connect to a network.

Q4. 50 is valid. 101 is an integer but outside the range. Text 'fifty' is the wrong type (erroneous). Validation checks rules; it cannot prove the pupil earned 50.

I Do / We Do pages · Guided practice

Q1. It performs a specific function rather than general-purpose computing.

You Do pages · Code First

```
clockMHz = 2
seconds = 3
millions = clockMHz * seconds
print(millions)
```

Expected result: 6

You Do pages · Independent questions

Q1. Clock speed, cache size, number of cores (any two).

Q2. Frequently needed instructions/data can be accessed faster, reducing waiting for RAM.

Q3. For example a washing-machine controller: built into the device for a dedicated control task.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: